

# 12. Applications of Linear Forms in Logarithms

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## 12.1 Introduction

A linear form in logarithms of algebraic numbers is an expression of the form

$$\beta_1 \log \alpha_1 + \cdots + \beta_n \log \alpha_n,$$

where the  $\alpha$ 's and the  $\beta$ 's denote complex algebraic numbers, and  $\log$  denotes any determination of the logarithm.

### 12.1.1 Lower Bounds

The first lower bound for such a sum was obtained in 1935 by Gel'fond [Gel] for the case  $n = 2$  of two logarithms. A giant step was made in 1966 by A. Baker [Bak1], who was able to deal with such a form for arbitrary  $n$ . Subsequently, many papers were published on this problem, by Baker, Fel'dman, etc., and this field is often called Baker's theory. We give only some references of works after 1990, see [Bak-Wus, BBGMS, BMS2, Lau, Lau-Mig-Nes, Mat, Wald1], and much more information can be found in the book by Waldschmidt [Wald2]. It is important to note that there are essentially two kinds of results: general estimates valid for any  $n$  as in [Bak-Wus, Mat, Wald1], and specific results for two logarithms as in [Lau, Lau-Mig-Nes], or for three logarithms as in [BBGMS, BMS2], which are crucial for the complete resolution of Diophantine equations.

Here we will consider only the case in which the  $\beta$ 's are in  $\mathbb{Z}$ , and they will be denoted by  $b_1, \dots, b_n$ . This is the only case which has applications to Diophantine equations. Also, in this chapter  $\log$  will always denote the principal determination of the complex logarithm.

It will be sufficient for us to give the general lower bound for linear forms in logarithms due to Matveev [Mat]. Let  $\mathbb{L}$  be a number field of degree  $D$ , let  $\alpha_1, \dots, \alpha_n$  be nonzero elements of  $\mathbb{L}$ , and let  $b_1, \dots, b_n$  be integers. Set

$$B = \max\{|b_1|, \dots, |b_n|\},$$

and

$$\Lambda^* = \alpha_1^{b_1} \cdots \alpha_n^{b_n} - 1.$$

We wish to bound  $|\Lambda^*|$  from below, assuming that it is nonzero. Since  $\log(1+x)$  is asymptotic to  $x$  as  $|x|$  tends to 0, our problem consists in bounding from below the “linear form in logarithms”

$$\Lambda = b_1 \log \alpha_1 + \cdots + b_n \log \alpha_n + b_{n+1} \log(-1),$$

where  $b_{n+1} = 0$  if  $\mathbb{L}$  is real, and  $|b_{n+1}| \leq nB$  otherwise. Although  $\Lambda$  and  $\Lambda^*$  are closely linked (in particular one vanishes if and only if the other does), it is useful to keep both, and this dual notation will be used in the sequel without further explanation.

Recall that we define the *absolute logarithmic height* of an algebraic number  $\alpha$  as follows.

**Definition 12.1.1.** *Let  $\mathbb{L}$  be a number field of degree  $D$ , let  $\alpha \in \mathbb{L}^*$  be of degree  $d \mid D$ , and let  $\sum_{0 \leq k \leq d} a_k X^k$  be its minimal primitive polynomial in  $\mathbb{Z}[X]$  with  $a_d \neq 0$ . We define the absolute logarithmic height  $h(\alpha)$  of  $\alpha$  by one of the two equivalent formulas*

$$\begin{aligned} h(\alpha) &= \frac{1}{d} \left( \log(|a_d|) + \sum_{1 \leq i \leq d} \max(\log(|\alpha_i|), 0) \right) \\ &= \frac{1}{D} \sum_{v \in \mathcal{P}(\mathbb{L})} \max(\log(|\alpha|_v), 0), \end{aligned}$$

where the  $\alpha_i$  are the conjugates of  $\alpha$  and  $\mathcal{P}(\mathbb{L})$  denotes the set of places of  $\mathbb{L}$  (see Definition 4.1.12).

It is immediate to check that these formulas are equivalent, hence do not depend on the number field  $\mathbb{L}$  containing  $\alpha$ , and that if  $\alpha = n/d \in \mathbb{Q}$  with  $\gcd(n, d) = 1$  then  $h(\alpha) = \max(\log(|n|), \log(|d|))$ , so this generalizes to algebraic numbers the usual notion of height that we have already used in Section 8.1.5.

Let  $A_1, \dots, A_n$  be real numbers such that

$$A_j \geq h'(\alpha_j) := \max\{D h(\alpha_j), |\log \alpha_j|, 0.16\}, \quad 1 \leq j \leq n.$$

We call  $h'$  the modified height (with respect to the field  $\mathbb{L}$ ). With this notation, the main result of Matveev [Mat] implies the following estimate.

**Theorem 12.1.2.** *Assume that  $\Lambda^*$  (defined above) is nonzero. We then have*

$$\log |\Lambda^*| > -3 \cdot 30^{n+4} (n+1)^{5.5} D^2 A_1 \cdots A_n (1 + \log D) (1 + \log nB).$$

Furthermore, if  $\mathbb{L}$  is real, we have

$$\log |\Lambda^*| > -1.4 \cdot 30^{n+3} n^{4.5} D^2 A_1 \cdots A_n (1 + \log D) (1 + \log B).$$

**Remark.** In several applications to Diophantine equations we need a better estimate in terms of  $B$ , as in the following result ([Wald1], [Wald2] Theorem. 9.1): with the above notation and assuming that the algebraic numbers  $\alpha_1, \dots, \alpha_n$  are multiplicatively independent and that  $b_n \neq 0$ , there exists a positive effective constant  $C(n)$ , which depends only on  $n$ , such that

$$\log |\Lambda| > -C(n) \cdot D^2 (\log D) A_1 \cdots A_n \log B',$$

where

$$B' = \max_{1 \leq j \leq n} \left\{ \frac{|b_n|}{A_j} + \frac{|b_j|}{A_n} \right\}.$$

Compared to Theorem 12.1.2, this gives an improvement especially when  $\alpha_n$  has a large height and  $|b_n|$  is small (see Section 12.10 for the application to Thue equations).

It is important to compare the above lower bound for  $|\Lambda|$  with the elementary lower bound obtained by a Liouville-type argument, which gives

$$\log |\Lambda| > -D(1 + |b_1| h(\alpha_1) + |b_2| h(\alpha_2) + \cdots + |b_n| h(\alpha_n)).$$

In this estimate the dependence with respect to  $D$  and to each  $h(\alpha_j)$  is better than in the theorem, *but* in the theorem the dependence on  $B$  is logarithmic, while it is linear in this elementary estimate. This makes all the difference, and this elementary estimate has no applications to Diophantine equations. Actually, for applications we do not need a lower bound that is logarithmic in terms of  $B$ : in many cases (but not in all; see Catalan's equation in Section 12.9) a result like the following would be sufficient: for any  $\varepsilon > 0$  there exists a positive constant  $C_\varepsilon$  such that

$$\log |\Lambda| > -\varepsilon B \quad \text{for } B > C_\varepsilon,$$

where  $C_\varepsilon$  does not depend of  $B$ , but on  $\alpha_1, \dots, \alpha_n$  and their logarithms. In practice, the best results for two or three logarithms depend on a term in  $\log^2 B$ .

### 12.1.2 Applications to Diophantine Equations and Problems

For applications to Diophantine problems, the strategy is the following. In a first step, various and often ad hoc algebraic manipulations associate to a “large” solution of the equation a “very small” value of a certain linear form in logarithms, which means that we have an upper bound for the values of this linear form corresponding to a solution of the equation. Comparing this upper bound with the lower bound provided by Theorem 12.1.2, we get an absolute upper bound  $M$  for the absolute values of the unknowns of our equation.

At this point, there are two main cases in which we can go from a bound for the unknowns to a complete solution of the equation.

*The first case* is that  $M$  is not too large. Then, using various methods including sieves, we find the complete list of solutions below  $M$ . For this to be possible, it is crucial to get a “reasonably small” value for  $M$ . Actually, its size is directly related to the size of the “numerical constant” that appears in Theorem 12.1.2, in other words the factor  $1.4 \cdot 30^{n+3} n^{4.5}$  occurring in the second estimate.

Many celebrated Diophantine equations lead to estimates of linear forms in two or three logarithms. In these cases, Theorem 12.1.2 gives numerical constants around  $10^{12}$  and  $10^{14}$ , respectively. Both are usually too large for practical applications. Fortunately, an alternative approach, developed by Waldschmidt, Laurent, and Mignotte, among others, yields much better numerical constants with, however, a worse dependence on  $B$  (or in  $B'$ ), namely with the factor  $\log B$  replaced by its square. Despite this worse dependence on  $B$ , the results from [Lau, Lau-Mig-Nes, BBGMS, BMS2] are, for many practical applications, better than Theorem 12.1.2.

*A second very important special case* is that the linear form constructed above has the following property: *only the coefficients  $b_i$  are unknown*. From the bound  $|b_i| \leq M$  obtained using the type of argument outlined above, one can derive a much smaller bound. Indeed, effective Diophantine approximation techniques (continued fractions, LLL algorithm) can be used to obtain a good lower bound for

$$\min_{\substack{|b_i| \leq M \\ (b_i)_{1 \leq i \leq n} \neq 0}} \left| \sum_{i=1}^n b_i x_i \right| ,$$

which can be used in place of the Baker-type estimate. Comparing this new bound with the upper bound, we get a value  $M'$  with  $|b_i| \leq M'$ ; we can repeat this process until no new improvement is obtained. The technicalities of this method have already been worked out in Section 2.3.5, and the method applied in Section 8.7.

### 12.1.3 A List of Applications

The list of applications given in this chapter may look unusual to experts. We want to begin with results for which the reduction

Diophantine problem  $\longrightarrow$  linear form in logarithms

is almost obvious. The chosen examples correspond to such reductions in increasing order of difficulty. This is the reason why the most important example of Thue equations is near the end of our list. A more classical presentation can be found in the book by Baker [Bak2], but our presentation has some similarities with that in the book by Shorey and Tijdeman [Sho-Tij].

## 12.2 A Lower Bound for $|2^m - 3^n|$

One of the simplest applications of linear forms in logarithms is to prove that  $|2^m - 3^n|$  tends to infinity with  $m + n$ , and, in addition, to give an explicit lower bound for this quantity.

Let  $n \geq 2$  be an integer and define  $m$  and  $m'$  by the conditions

$$2^{m'} < 3^n < 2^{m'+1} \quad \text{and} \quad |3^n - 2^m| = \min\{3^n - 2^{m'}, 2^{m'+1} - 3^n\}.$$

Then

$$|2^m - 3^n| < 2^m, \quad (m-1)\log 2 < n\log 3 < (m+1)\log 2,$$

and the problem of finding a lower bound for  $|2^m - 3^n|$  clearly reduces to this special case.

Thus, consider the “obvious” linear form

$$\Lambda^* = 3^n 2^{-m} - 1.$$

Applying Matveev’s theorem we get

$$\log |\Lambda^*| > -c_0 (1 + \log m),$$

where it is easy to verify that we can take  $c_0 = 5.87 \times 10^8$ .

This implies the following estimate.

**Theorem 12.2.1.** *Let  $m$  and  $n$  be any strictly positive integers, then*

$$|2^m - 3^n| > 2^m (em)^{-5.87 \times 10^8}.$$

More generally, if  $S$  denotes a finite set of prime numbers and if  $(x_j)_{j \geq 1}$  is the increasing sequence of all integers whose prime divisors belong to  $S$ , then it follows from Theorem 12.1.2 that

$$|x_{j+1} - x_j| \geq x_j (\log x_j)^{-c},$$

where the constant  $c$  can be explicitly computed in terms of the prime numbers in  $S$  (see [Sho-Tij]).

Theorem 12.2.1 enables us to find the list of all powers of 3 that increased by 5 give a power of 2.

**Corollary 12.2.2.** *The only integer solutions to the Diophantine equation*

$$2^m - 3^n = 5$$

*are  $(m, n) = (3, 1)$  and  $(m, n) = (5, 3)$ , which correspond respectively to  $8 - 5 = 3$  and to  $32 - 27 = 5$ .*

*Proof.* Applying the above theorem to this equation we get

$$5 > 2^m \cdot (em)^{-5.87 \times 10^8},$$

which implies

$$\log 5 > m \log 2 - 5.87 \times 10^8 (1 + \log m),$$

so that  $m < 2.1 \times 10^{10}$  and  $n < m \log 2 / \log 3 < 1.4 \times 10^{10}$ .

Moreover, the relation  $2^m - 3^n = 5$  implies

$$\left| m - n \frac{\log 3}{\log 2} \right| < \frac{5}{\log 2} 3^{-n}.$$

This identity has a first consequence: since  $(5/\log 2) 3^{-n} < \frac{1}{2^n}$  for  $n \geq 4$ , we see that if  $(m, n)$  is a solution to our problem with  $n \geq 4$  then  $m/n$  is a convergent of the continued fraction expansion of  $\xi := \log 3 / \log 2$ .

But we can also notice the following: for  $n < N = 1.4 \times 10^{10}$ , the smallest value of  $|m - n\xi|$  is obtained for the largest convergent of the continued fraction expansion of  $\xi$  with denominator less than  $N$ . The computation of this expansion shows that

$$\frac{5}{\log 2} 3^{-n} > \left| m - n \frac{\log 3}{\log 2} \right| > 10^{-11}, \quad \text{for } 0 < n < 1.4 \times 10^{10}.$$

Comparing those two estimates, we see that if  $(m, n)$  is a solution of our problem then  $n \leq 24$ . A trivial verification in the range  $1 \leq n \leq 24$  (or a less trivial verification using the fact that for  $n \geq 4$ ,  $m/n$  must be a convergent of  $\xi$ , hence either  $19/12$  or  $8/5$ ) completes the proof.  $\square$

Notice that we are in effect solving the Diophantine inequality  $0 \leq 2^m - 3^n \leq 5$  in all steps except the final verification.

More generally, we have the following result (see [Ben3]):

**Theorem 12.2.3 (Bennett).** *For given nonzero integers  $a$ ,  $b$ , and  $c$  the equation  $a^m - b^n = c$  has at most two integer solutions.*

To conclude this section, we indicate how a similar, generalized strategy can be used to solve the equation  $0 \leq u - v \leq X$ , where  $u$  and  $v$  are integers with all their prime factors in a given set  $\{p_1, \dots, p_r\}$ . Write  $u = \prod_{i=1}^r p_i^{u_i}$ ,  $v = \prod_{i=1}^r p_i^{v_i}$ . If we restrict to primitive solutions, i.e., with  $(u, v) = 1$  (non-primitive solutions can be easily enumerated as multiples of primitive ones), we can assume that for all  $i$ , at least one of  $u_i, v_i$  is zero.

We have

$$\Lambda^* := \prod_{i=1}^r p_i^{u_i - v_i} - 1 \leq Xv^{-1},$$

which implies that

$$0 \leq \Lambda := \sum_{i=1}^r (u_i - v_i) \log p_i \leq Xv^{-1}.$$

Put  $m_i = u_i - v_i$ , and  $M = \max_i |m_i|$ . Assume further that the  $p_i$  are sorted in increasing order, i.e.,  $p_1 < p_2 < \dots < p_r$ .

We need to rewrite these upper bounds in terms of  $M$  rather than  $v$ . Either  $M = \max_i |v_i|$ , in which case we have  $v \geq p_1^M$ , or  $M = \max_i |u_i|$ , in which case we have  $v \geq u - X \geq u/2 \geq p_1^M/2$  if  $u \geq 2X$ ; hence  $Xv^{-1} \leq 2X \exp(-M \log p_1)$ .

Comparing this upper bound with the lower bound coming from Matveev's theorem, we deduce

$$M \log p_1 \leq \log(2X) + 1.4 \cdot 30^{r+3} r^{4.5} \left( \prod_{i=1}^r \log p_i \right) (1 + \log M).$$

We obtain an upper bound on  $M$  by using, for instance, the following easy lemma, due in this form to Pethő and de Weger:

**Lemma 12.2.4.** *Let  $B$  be a nonnegative integer such that*

$$\alpha \log B + \beta \geq \gamma B.$$

*If  $\alpha \geq e\gamma$ , we have*

$$B \leq \frac{2}{\gamma} \left( \alpha \log \frac{\alpha}{\gamma} + \beta \right).$$

*Proof.* Exercise for the reader. □

Once combined with the reduction technique described in Section 2.3, this leads to the following:

**Algorithm 12.2.5 (Find all  $0 \leq u - v \leq X$  with given prime factors)**

Let  $p_1 < \dots < p_r$  be primes and  $X$  a nonnegative real number. This algorithm outputs all the solutions of the Diophantine equation  $0 \leq u - v \leq X$ , with  $u, v$  integers having all their prime factors in the set  $\{p_1, \dots, p_r\}$ .

1. [Compute constants for Baker's bound] Compute the constants involved in the upper and lower bounds above:

$$\lambda_1 = \log p_1, \quad \lambda_3 = 1.4 \cdot 30^{r+3} r^{4.5} \left( \prod_{i=1}^r \log p_i \right), \quad \lambda_2 = \lambda_3 + \log(2X).$$

2. [Derive a large upper bound on  $M$ ] Compute the bound

$$M = \frac{2}{\lambda_3} \left( \lambda_1 \log \frac{\lambda_1}{\lambda_3} + \lambda_2 \right).$$

3. [Reduction step, initialization] Let  $C$  be a nonnegative real number somewhat larger than  $(rM)^r$ , say  $100(rM)^r$ .
4. [Reduction step] Apply the LLL algorithm to the lattice generated by the columns of the following matrix:

$$\begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \\ [C \log p_1] & [C \log p_2] & \dots & [C \log p_{r-1}] & [C \log p_r] \end{pmatrix}$$

Let  $l_0$  be the  $L^2$ -norm of the first vector of the reduced basis.

5. [Reduction step, new upper bound] If  $l_0 < 2^{(r-1)/2} \sqrt{(r^2/4 + (r-1))M}$ , put  $C \leftarrow 100C$  and go to Step 4. Otherwise, set

$$M' = -\frac{1}{\log p_1} \log \left( \frac{\sqrt{2^{1-r} l_0^2 - (r-1)M} - rM/2}{2CX} \right).$$

If  $\lfloor M' \rfloor < M$ , set  $M \leftarrow \lfloor M' \rfloor$  and go to Step 3.

6. [Final enumeration] Set  $\mathcal{S}_0 = \emptyset$ . For all  $2r$ -tuples  $((u_i)_{1 \leq i \leq r}, (v_i)_{1 \leq i \leq r})$  with  $u_i v_i = 0$  and  $u_i, v_i \leq M$ , set  $u = \prod_{i=1}^r u_i, v = \prod_{i=1}^r v_i$ , check whether  $0 \leq u - v \leq X$ , and if this is the case, add  $(u, v)$  to the set  $\mathcal{S}_0$  of the solutions.
7. [Small solutions] Look for solutions with  $u \leq 2C$  and add them to  $\mathcal{S}_0$ .
8. [Nonprimitive solutions] Set  $\mathcal{S} = \emptyset$ . For all solutions  $(u, v)$  in  $\mathcal{S}_0$  and all primes  $p$  of  $\{p_1, \dots, p_r\}$ , do the following:  $(u_0, v_0) \leftarrow (u, v)$ ; while  $|u_0 - v_0| \leq C$ , add  $(u_0, v_0)$  to  $\mathcal{S}$  and do  $(u_0, v_0) \leftarrow (pu_0, pv_0)$ .
9. [Terminate] Return  $\mathcal{S}$ .

**Remarks.** (1) The constant 100 involved in the choice of  $C$  is somewhat arbitrary and should be tuned somehow.

- (2) The algorithm given above is somewhat naïve, especially regarding the final enumeration.
- (3) This method can again be generalized to solve the equation  $A + B = C$  when  $A, B, C$  have all their prime factors in the set  $\{p_1, \dots, p_r\}$ , using non-Archimedean arguments (see Section 12.12). This approach has been used, in particular, to obtain “worst cases” for the *abc* conjecture (Conjecture 14.6.4). See [DeW1] for extensive computations on this problem.

### 12.3 Lower Bounds for the Trace of $\alpha^n$

Let  $\alpha$  be a nonzero algebraic integer of degree  $d > 1$  that is not a root of unity. Assume that its conjugates  $\alpha = \alpha_1, \dots, \alpha_d$  (in the field  $\mathbb{C}$ ) satisfy



$$|\alpha_1| \geq |\alpha_2| > |\alpha_3| \geq \cdots \geq |\alpha_d|.$$

Note that Kronecker's theorem (Corollary 3.3.10) implies that  $|\alpha| > 1$ , and that the trace of  $\alpha^n$  is by definition

$$\text{Trace}(\alpha^n) = \alpha_1^n + \alpha_2^n + \cdots + \alpha_d^n.$$

The purpose of this section is to show how one derives from Theorem 12.1.2 a lower bound for  $\text{Trace}(\alpha^n)$ .

In the (trivial) case  $|\alpha_1| > |\alpha_2|$  we have

$$|\alpha_1|^n - (d-1)|\alpha_2|^n \leq |\text{Trace}(\alpha^n)| \leq |\alpha_1|^n + (d-1)|\alpha_2|^n,$$

so  $|\text{Trace}(\alpha^n)| \approx |\alpha_1|^n$ . But suppose now that  $|\alpha_1| = |\alpha_2|$  and that we are not in the “degenerate” case  $\alpha_2 = -\alpha_1$ . Then  $\alpha_2$  is the complex conjugate of  $\alpha_1$ . Set  $\alpha = \rho e^{i\varphi}$ , with  $\rho > 0$ , and observe that

$$\alpha_1^n + \alpha_2^n = \rho^n (e^{in\varphi} + e^{-in\varphi}) = 2\rho^n \cos(n\varphi).$$

This elementary formula shows that obtaining a lower bound for  $|\alpha_1^n + \alpha_2^n|$  is exactly equivalent to obtaining a lower bound for

$$\Lambda_1 := ni\varphi - ki\pi,$$

where  $i = \sqrt{-1}$  and  $k$  is an integer. Our problem thus becomes a Diophantine approximation problem. More precisely,  $\Lambda_1$  is a *linear form in the logarithms of algebraic numbers with integer coefficients*. Indeed,

$$i\varphi = \log(\alpha/|\alpha|), \quad i\pi = \log(-1),$$

where  $\alpha/|\alpha|$  and  $-1$  are algebraic numbers. Applying Theorem 12.1.2, it is easy to see that

$$\log |\Lambda_1| \geq -c_1 \log n,$$

where  $c_1$  is a positive real number that depends only on  $\alpha$ . We immediately conclude that in the present situation,

$$|\text{Trace}(\alpha^n)| \geq 0.5 |\alpha|^n n^{-c_1}, \quad \text{for } n > c_2,$$

for some positive constants  $c_1$  and  $c_2$  that depend only on  $\alpha$ . Here and in the sequel all the constants  $c_i$  are positive, effective, and indeed easy to compute explicitly, and we will indicate their dependence in terms of the parameters.

More generally, we can prove the following result.

**Theorem 12.3.1.** *Assume that  $a$  and  $\alpha$  are nonzero algebraic numbers with  $\alpha$  nonreal. Then*

$$|a\alpha^n + \bar{a}\bar{\alpha}^n| > |\alpha|^n n^{-c_3},$$

for  $n > c_4$ , where  $c_3$  and  $c_4$  depend only on  $a$  and  $\alpha$ .

It is clear that such a result has applications to linear recurrent sequences; for this question, see [Sho-Tij] for more details.

## 12.4 Pure Powers in Binary Recurrent Sequences

Consider first the example of Fibonacci and Lucas numbers. Recall that these two sequences  $(F_n)$  and  $(L_n)$  are defined respectively by

$$F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}, \quad L_n = \alpha^n + \beta^n, \quad \text{where} \quad \alpha = \frac{1 + \sqrt{5}}{2} \text{ and } \beta = \frac{1 - \sqrt{5}}{2},$$

so that for  $n \geq 0$ ,

$$\begin{aligned} (F_n) &= 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \dots \text{ and} \\ (L_n) &= 2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, 199, 322, \dots \end{aligned}$$

Suppose that  $F_n = y^p$  is a pure power. Then, we have

$$\alpha^n - \sqrt{5}y^p = O(\alpha^{-n}),$$

so that

$$\Lambda_2 := n \log \alpha - p \log y - \log \sqrt{5} = O(\alpha^{-2n}) = O(y^{-2p}).$$

There exist integers  $k$  and  $r$  such that  $n = kp + r$  with  $|r| \leq p/2$ , so that we have

$$\Lambda_2 = p \log \left( \frac{\alpha^k}{y} \right) + r \log \alpha - \log \sqrt{5},$$

which is a linear form in three logarithms. If we apply the above theorem of Matveev, we get

$$\log |\Lambda_2| \geq -c_5 \log y \log p.$$

Comparing both estimates of  $|\Lambda_2|$ , we see that the exponent  $p$  is bounded. More precisely, Matveev's theorem above implies  $p < 3 \times 10^{13}$ , but a special estimate for linear forms in three logarithms proved in [BMS1] implies the sharper upper bound  $p < 2 \times 10^8$ , a range that is suitable for computer calculations. For Lucas numbers, a similar study leads to a linear form in two logarithms—a much better situation—and an application of [Lau-Mig-Nes] leads to  $p < 300$  if  $L_n = y^p$ . In the case of Fibonacci numbers, the application of the “modular method” (see Chapter 15) allows us to prove that  $r = \pm 1$ , for any prime  $p$  in the above range. Thus

$$\Lambda_2 = p \log \left( \frac{\alpha^k}{\log y} \right) + \log \left( \frac{\alpha^{\pm 1}}{\sqrt{5}} \right)$$

is a linear form in two logarithms (!), and an application of [Lau-Mig-Nes] now leads to  $p < 733$ .

Coming back to a more general situation we have the following result.

**Theorem 12.4.1.** *Suppose that  $(u_n)$  is a sequence of integers of the form*

$$u_n = a\alpha^n + O(|\alpha|^{\theta n}), \quad \text{with } 0 < \theta < 1,$$

*where  $a$  and  $\alpha$  are nonzero algebraic numbers, with  $|\alpha| > 1$  and  $\theta$  fixed, and that  $u_n - a\alpha^n \neq 0$  for all  $n$ . The equation*

$$u_n = y^p, \quad u_n \notin \{0, \pm 1\},$$

*implies that  $p < c_6$ , where the upper bound  $c_6$  depends only on  $a$ ,  $\alpha$ ,  $\theta$ , and on the implicit constant in the above  $O$ .*

This result evidently applies in particular to nondegenerate linear binary recurrent sequences with real “roots” and, more generally, to linear recurrent sequences with exactly one “dominant root,” and moreover for which this dominant root is simple. See again [Sho-Tij] for more details.

Note that it has been recently proved in [BMS1] that the only perfect powers in the Fibonacci and Lucas sequences are exactly the powers that appear in the previous list, in other words 0, 1, 8, and 144 for the Fibonacci numbers, and 1 and 4 for the Lucas numbers.

## 12.5 Greatest Prime Factors of Terms of Some Recurrent Sequences

As in the previous section consider a sequence of nonzero integers  $u_n$  such that

$$u_n = a\alpha^n + O(|\alpha|^{\theta n}), \quad 0 < \theta < 1, \quad u_n \neq a\alpha^n,$$

where  $a$  and  $\alpha$  are nonzero algebraic numbers with  $|\alpha| > 1$ , and where  $\theta$  is fixed. Let  $p_1 = 2, p_2 = 3, p_3 = 5, \dots$  be the sequence of prime numbers in increasing order. Assume that the largest prime factor of  $u_n$  is equal to  $p_k$ , in other words that

$$u_n = p_1^{r_1} \cdots p_k^{r_k},$$

with  $r_k > 0$ . Set

$$\Lambda_3 = n \log \alpha + \log a - r_1 \log p_1 - \cdots - r_k \log p_k.$$

The definition of  $u_n$  implies that

$$\log |\Lambda_3| \leq -c_7 n,$$

where  $c_7$  is a positive constant that depends only on  $a$ ,  $\alpha$ ,  $\theta$ , and on the implicit constant in the above  $O$ .

In the other direction, Theorem 12.1.2 implies that

$$\log |\Lambda_3| \geq -c_8 (\log p_k)^k \log n,$$

where  $c_8$  is a constant depending only on  $a$  and  $\alpha$ . Comparing both estimates and using the estimate  $p_k \sim k \log k$ , which is equivalent to the prime number theorem, we obtain the following result.

**Theorem 12.5.1.** *Let  $(u_n)$  be a sequence of nonzero integers of the form*

$$u_n = a\alpha^n + O(|\alpha|^{\theta n}), \quad 0 < \theta < 1,$$

*where  $a$  and  $\alpha$  are nonzero algebraic numbers, with  $|\alpha| > 1$  and  $\theta$  fixed, and assume that  $u_n - a\alpha^n \neq 0$  for all  $n$ . Let  $p_1 = 2, p_2 = 3, p_3 = 5, \dots$  be the sequence of prime numbers, and suppose that the largest prime factor of  $u_n$  is equal to  $p_k$ . Then*

$$k \geq c_9 \log n / \log \log \log n,$$

*where  $c_9$  is a positive constant that depends only on  $a, \alpha, \theta$ , and on the implicit constant in the above  $O$ .*

## 12.6 Greatest Prime Factors of Values of Integer Polynomials

Let  $f(X)$  be an irreducible polynomial in  $\mathbb{Z}[X]$  of degree greater than or equal to 2, and let  $x$  be a strictly positive integer. Using Baker's theory, it is possible to give a lower bound for the greatest prime factor of  $f(x)$ . Take for instance  $f(X) = X(X-1)$  and, with the same notation as in the previous section, write

$$x(x-1) = p_1^{r_1} \cdots p_k^{r_k},$$

with  $r_k > 0$ . Then, for suitable  $\varepsilon_i$  in  $\{\pm 1\}$ , we get

$$|p_1^{\varepsilon_1 r_1} \cdots p_k^{\varepsilon_k r_k} - 1| \leq \frac{1}{x-1}.$$

Since  $p_j \sim j \log j$  we deduce from Theorem 12.1.2 that there exists an absolute positive constant  $c_{10}$  such that

$$\log x \leq c_{10}^{k \log \log k} \log \log x.$$

This implies that

$$p_k \sim k \log k \gg \log \log x \frac{\log \log \log x}{\log \log \log \log x}.$$

A similar result holds for any irreducible polynomial in  $\mathbb{Z}[X]$  of degree greater than or equal to 2.

## 12.7 The Diophantine Equation $ax^n - by^n = c$

In this section we consider the exponential Diophantine equation

$$ax^n - by^n = c,$$

where  $a$ ,  $b$ , and  $c$  are nonzero fixed integers, with  $a$  and  $b$  strictly positive, and where  $x$ ,  $y$ , and  $n$  are unknowns. If for some exponent  $n$  there exists a solution  $(x, y)$  with  $|y| > 1$  then

$$\Lambda_4 = \log |a/b| - n \log |x/y| = O(|y|^{-n}).$$

In the other direction, Theorem 12.1.2 above implies that

$$\log |\Lambda_4| \geq -c_{11} \log |y| \cdot \log n.$$

Comparing both estimates we see that

$$n < c_{12},$$

where  $c_{12}$  depends on  $a$ ,  $b$ , and  $c$ .

We give an explicit version of this result, established in [Mig]. Its proof does not depend on Theorem 12.1.2, but on [Lau-Mig-Nes]. Actually, Theorem 12.1.2 does not include all known refinements. In particular, as first noticed by Shorey, it can be considerably improved when  $\alpha_1, \dots, \alpha_n$  are real numbers all very close to 1: roughly speaking, the product of the  $A_i$ 's can then be replaced by their sum. This is precisely what is used in the proof of the next result.

**Theorem 12.7.1.** *Assume that the exponential Diophantine inequality*

$$|ax^n - by^n| \leq c, \quad \text{with } a, b, c \in \mathbb{Z}_{>0} \text{ and } a \neq b,$$

*has a solution in strictly positive integers  $x$  and  $y$  with  $\max\{x, y\} > 1$ . Then*

$$n \leq \max \left\{ 3 \log(1.5 |c/b|), 7400 \frac{\log A}{\log(1 + (\log A)/\log |a/b|)} \right\},$$

*where  $A = \max\{a, b, 3\}$ .*

It is remarkable that when  $|c|$  is very small and the ratio  $|a/b|$  is very close to 1, the upper bound given by Theorem 12.7.1 is absolute; in other words, it does not depend on  $a$ ,  $b$ , and  $c$ . For instance, if there exist strictly positive integers  $n \geq 3$ ,  $b$ ,  $c$ ,  $x$ , and  $y$  with  $\max\{x, y\} > 1$  and  $b \geq c^2$ , satisfying

$$|(b+c)x^n - by^n| \leq c,$$

then  $n$  is less than some *absolute* constant.

Using Padé approximations, Bennett and de Weger [Ben-deW] improved the previous result, and ultimately Bennett [Ben1] obtained the following definitive result for  $c = \pm 1$  (the case  $n = 3$  is essentially Skolem's Theorem 6.4.30).

**Theorem 12.7.2 (Bennett).** *If  $n \geq 3$  the equation*

$$|ax^n - by^n| = 1, \quad \text{with } a, b \in \mathbb{Z}_{>0},$$

*has at most one solution in strictly positive integers  $x$  and  $y$ .*

For instance, we thus know the complete list of solutions to the parametric family of equations  $(b+1)x^n - by^n = \pm 1$ .

Note that such equations are called *Thue equations*, and will be considered in more detail below.

## 12.8 Simultaneous Pell Equations

Consider the so-called simultaneous Pell equations

$$x^2 - ay^2 = 1, \quad x^2 - bz^2 = 1,$$

where  $a \geq 2$  and  $b \geq 2$  are distinct squarefree integers, in the integer unknowns  $x$ ,  $y$ , and  $z$ .

### 12.8.1 General Strategy

Let  $\varepsilon$  and  $\eta$  be the fundamental units of norm 1 of the real quadratic orders  $\mathbb{Z}[\sqrt{a}]$  and  $\mathbb{Z}[\sqrt{b}]$ , respectively. Changing if necessary  $x$ ,  $y$ , and  $z$  into their opposites we then have the relations

$$x + y\sqrt{a} = \varepsilon^m \quad \text{and} \quad x + z\sqrt{b} = \eta^n,$$

for suitable nonnegative integers  $m$  and  $n$ . In particular,

$$2x = \varepsilon^m + \varepsilon^{-m} = \eta^n + \eta^{-n}.$$

Consider the linear form

$$\Lambda_5 = m \log \varepsilon - n \log \eta,$$

which satisfies

$$\Lambda_5 = O(\varepsilon^{-m} + \eta^{-n}).$$

It follows from Theorem 12.1.2 that  $m$  and  $n$  are bounded (we leave the proof as an exercise for the reader). Consequently, the above system of two simultaneous Pell equations has only a finite number of solutions  $(x, y, z)$ .

As an example of a problem that leads to a system of simultaneous Pell equations consider the following. A Diophantine  $m$ -tuple is a set of  $m$  integers such that the product of any two of them, increased by 1, is a perfect square. A famous example is the quadruple  $\{1, 3, 8, 120\}$  found by Fermat. In 1969,

Baker and Davenport [Bak-Dav] proved that the set  $\{1, 3, 8, 120\}$  cannot be extended to a Diophantine quintuple. Actually, they established a stronger result: the only strictly positive integer  $t$  such that  $\{1, 3, 8, t\}$  is a Diophantine quadruple is 120. It is an easy exercise to see that this problem reduces to the study of the simultaneous equations

$$3x^2 - 2 = y^2 \quad \text{and} \quad 8x^2 - 7 = z^2.$$

They have only the “trivial” solution  $(x, y, z) = (\pm 1, \pm 1, \pm 1)$  and the solution  $(x, y, z) = (\pm 11, \pm 19, \pm 31)$ , which corresponds to  $t = 120$ . This was the first Diophantine problem completely solved by Baker’s theory and it involved computations with 1000 decimal digits, something very new at that time.

Another problem that can be reduced to a system of two simultaneous Pell equations is the so-called cannonball equation  $6y^2 = x(x+1)(2x+1)$ . Indeed, by studying the factors of  $x$ ,  $x+1$ ,  $2x+1$  modulo squares and elementary congruence arguments, we find that either  $x = u^2$ ,  $x+1 = 2v^2$ ,  $2x+1 = 3w^2$  or  $x = 6u^2$ ,  $x+1 = v^2$ ,  $2x+1 = w^2$  for some integers  $(u, v, w)$ . Both cases lead to a system of simultaneous Pell equations. Note that we have already solved this problem by completely elementary methods in Section 6.8.2, and that it can also be solved using the methods of Section 8.7, which also rely on Baker-type arguments, involving linear forms in elliptic logarithms instead of ordinary logarithms.

### 12.8.2 An Example in Detail

We now study a particular example in detail. Consider the simultaneous Pell equations

$$x^2 - 2y^2 = x^2 - 3z^2 = 1.$$

It is easy to see that if  $x > 1$  is a solution then there exist strictly positive integers  $m$  (with  $m$  even) and  $n$  such that

$$u_m = v_n, \quad \text{where} \quad u_j = (\alpha^j + (-\alpha)^{-j})/2 \quad \text{and} \quad v_k = (\beta^k + \beta^{-k})/2,$$

with

$$\alpha = 1 + \sqrt{2}, \quad \beta = 2 + \sqrt{3}.$$

If we assume  $m, n \neq 0$ , we obtain

$$|\alpha^m - \beta^n| = |\alpha^{-m} - \beta^{-n}| \leq 1,$$

so that in fact

$$|\alpha^{-m} - \beta^{-n}| \leq \alpha^{-m} \beta^{-n}.$$

From this, we deduce that

$$|\alpha^m \beta^{-n} - 1| \leq \alpha^{-m} \beta^{-2n}.$$

If  $n \neq 0$ , we have  $|\alpha^m - \beta^n| \leq \alpha^m/2$ , so that the upper bound can be replaced by  $4\alpha^{-3m}$ , and we can see that  $m \geq n$ . Theorem 12.1.2 (with  $D = 4$ ) shows that

$$\log |\alpha^m \beta^{-n} - 1| \geq -1.37 \times 10^{12}(1 + \log m).$$

Comparing this with the upper bound, we obtain  $m < 1.63 \times 10^{13}$ .

To obtain a more practical bound on  $m$ , we need to work with  $\Lambda = m \log \alpha - n \log \beta$  rather than with  $\Lambda^*$ . An upper bound for  $\Lambda$  can be obtained, for instance, by means of the following elementary lemma whose proof is left to the reader:

**Lemma 12.8.1.** *If  $x \in \mathbb{C}$  is such that  $|x - 1| \leq 1/2$ , then  $|\log x| \leq 2|x - 1| \log 2$ .*

We obtain

$$|m \log \alpha - n \log \beta| \leq 8\alpha^{-3m} \log 2$$

as soon as  $4\alpha^{-3m} < 1/2$ , i.e.,  $m \geq 1$ .

Thus, as in the case of the equation  $2^n - 3^m = 5$ , we have to consider the continued fraction expansion of some quotient of two logarithms, here of  $\xi = \log \alpha / \log \beta$ . The computation of this continued fraction expansion shows that

$$\left| n - m \frac{\log \alpha}{\log \beta} \right| > 3.91 \times 10^{-14} \quad \text{when } m \leq 1.63 \times 10^{13}.$$

We thus deduce that  $1 \leq m \leq 12$ . Finally, a trivial computation shows that the only solution for  $u_m = v_n$  is  $(m, n) = (3, 2)$ . However, this does not correspond to a solution to our problem since  $m$  is odd.

To conclude, we have thus proved that the only strictly positive solution to the equations

$$|x^2 - 2y^2| = x^2 - 3z^2 = 1$$

is  $(x, y, z) = (7, 5, 4)$ .

### 12.8.3 A General Algorithm

We can formalize this procedure as an algorithm. First, we need a technical lemma, which will give us both a bound on  $\Lambda^* = \gamma \epsilon^n \delta^{-1} \eta^{-m}$ , to which we want to apply Matveev's theorem, and on  $\Lambda = \log \gamma \delta^{-1} + n \log \epsilon - m \log \eta$ , which we shall use in the reduction process.

**Lemma 12.8.2.** *Let  $\epsilon > 1, \eta > 1, \gamma$ , and  $\delta$  be real numbers. Under the assumptions*

$$|\gamma \epsilon^n - \delta \eta^m| \leq C \cdot \max(\epsilon^{-n}, \eta^{-m}), \quad n \neq 0, \quad \text{and}$$

$$m > M_0 := \frac{1}{\log \eta} \max \left( \log \frac{C|\gamma|}{|\delta|}, \frac{1}{2} \log \left\{ C \max \left( \frac{1}{2|\delta|}, \frac{1}{|\gamma|} \right) \right\} \right),$$

we have

$$|\log \gamma \delta^{-1} + n \log \epsilon - m \log \eta| \leq 2C(\log 2) \max(\delta^{-1}, 2\gamma^{-1}) \eta^{-2m}.$$



*Proof.* Since  $n, m > 0$ , we have  $|\gamma\varepsilon^n - \delta\eta^m| \leq C$ , hence  $|\varepsilon^n| \leq C + |\delta\gamma^{-1}\eta^m| \leq 2|\delta\gamma^{-1}\eta^m|$  for  $m > \log(C|\gamma\delta^{-1}|)/\log\eta$ . As a consequence, we have

$$\left| \frac{\gamma\varepsilon^n}{\delta\eta^m} - 1 \right| \leq C \max(\delta^{-1}, 2\gamma^{-1})\eta^{-2m}.$$

Since  $m > M_0$ , the left-hand side is at most  $1/2$  and we can apply Lemma 12.8.1.  $\square$

Two cases can occur in the reduction step: either we are in the homogeneous case (as above), and we can use elementary properties of continued fractions; or we are in the inhomogeneous case (the general case in which the right-hand side is nontrivial) and we can also use continued fractions as in the following lemma.

**Lemma 12.8.3 (Baker–Davenport).** *Let  $x_0, x_1, x_2$  be real numbers,  $b_1$  an integer with  $|b_1| \leq B$ , and  $Q$  a nonnegative integer. Then*

$$d(x_0 + b_1x_1, \mathbb{Z}) \geq \frac{1}{Q} \{d(Qx_0, \mathbb{Z}) - Bd(Qx_1, \mathbb{Z})\}.$$

*Proof.* This follows from the chain of inequalities

$$\begin{aligned} Qd(x_0 + b_1x_1, \mathbb{Z}) &\geq d(Qx_0 + Qb_1x_1, \mathbb{Z}) \\ &\geq d(Qx_0, \mathbb{Z}) - d(Qb_1x_1, \mathbb{Z}) \\ &\geq d(Qx_0, \mathbb{Z}) - Bd(Qx_1, \mathbb{Z}). \end{aligned}$$

$\square$

This lemma should be applied with  $Q$  the denominator of a convergent of the continued fraction expansion of  $x_1$ , with  $Q \approx \kappa B$ , since in that case, we expect  $Bd(Qx_1, \mathbb{Z}) \approx \kappa^{-1}$ . We obtain the following general algorithm:

**Algorithm 12.8.4 (Simultaneous Pell equations)** Let  $a_0, a_1, b_0, b_1$  be integers such that  $\sqrt{a_0/b_0}$  is not a rational number. This algorithm gives the list of solutions of the system  $x^2 - a_0y^2 = a_1, x^2 - b_0z^2 = b_1$ .

1. [Algebraic precomputations] Compute fundamental units  $\varepsilon$  of  $\mathbb{Q}(\sqrt{a_0})$  and  $\eta$  of  $\mathbb{Q}(\sqrt{b_0})$ ; we shall assume that they are chosen such that  $\varepsilon, \eta > 1$ . Compute two sets  $\mathcal{S}_0$  and  $\mathcal{S}_1$  of inequivalent solutions in algebraic integers of the norm equations  $N_{\mathbb{Q}(\sqrt{a_0})/\mathbb{Q}}(\gamma) = a_1, N_{\mathbb{Q}(\sqrt{b_0})/\mathbb{Q}}(\delta) = b_1$ . The following steps should be executed for all  $(\gamma, \delta) \in \mathcal{S}_0 \times \mathcal{S}_1$ .
2. [Compute Baker's bound] Compute the constants  $M_0$ ,

$$\begin{aligned}\lambda_0 &= a_1|\gamma|^{-1} + b_1|\delta|^{-1}, \quad \lambda_1 = 2.94 \times 10^{10} h'(\varepsilon) h'(\eta) h'(\delta/\gamma), \\ \lambda'_2 &= C \max(\delta^{-1}, 2\gamma^{-1}), \quad \lambda_2 = \lambda'_2 + \lambda_1, \quad \lambda_3 = 2 \log \eta, \quad \text{and} \\ M &= \frac{2}{\lambda_3} \left( \lambda_1 \log \frac{\lambda_1}{\lambda_3} + \lambda_2 \right).\end{aligned}$$

3. [Reduction, choice of the parameter] If  $\delta = \gamma$ , go to Step 5. Otherwise, choose  $C$  somewhat larger than  $M$ , say  $C = 100M$ .
4. [Reduction, inhomogeneous case] Compute the largest convergent  $p/q$  of the real number  $\log \eta / \log \varepsilon$  for which  $q \leq C$ . If

$$D := d(q \log \gamma \delta^{-1} / \log \varepsilon, \mathbb{Z}) - Md(q \log \eta / \log \varepsilon, \mathbb{Z}) < 0,$$

set  $M \leftarrow 100M$  and go to Step 4. Otherwise, compute

$$M' = -\frac{\log(D/2q\lambda'_2 \log 2)}{2 \log \eta}.$$

If  $\lfloor M' \rfloor < M$ , set  $M \leftarrow \lfloor M' \rfloor$  and go to Step 3; otherwise, go to Step 6.

5. [Reduction, homogeneous case] Compute the largest convergent  $p/q$  of the real number  $\log \eta / \log \varepsilon$  with  $q \leq M$ ,

$$D = |q \log \eta - p \log \varepsilon|, \quad \text{and} \quad M' = -\frac{\log \{D/(2q\lambda'_2 \log 2)\}}{2 \log \eta}.$$

If  $M' < M$ , set  $M \leftarrow M'$  and go to Step 5.

6. [Enumeration] Initialize  $\mathcal{S}$  to  $\emptyset$ . For each  $m < \max(M_0, M)$ , compute  $x = (\delta \eta^m + a_1 \delta^{-1} \eta^{-m})/2$ ; put  $y = \sqrt{(x^2 - a_1)/a_0}$ ,  $z = \sqrt{(x^2 - b_1)/b_0}$ . If  $y$  and  $z$  are rational integers, add  $(x, y, z)$  to  $\mathcal{S}$ .
7. [Terminate] Return  $\mathcal{S}$ .

**Remarks.** (1) In practice, the case  $\delta = \gamma$  should be treated via forms in 2 logarithms rather than via Matveev's bound.  
 (2) The enumeration step should be optimized, especially if  $\max(M_0, M)$  is large. For instance, one could use congruence conditions on  $m$ .

## 12.9 Catalan's Equation

Catalan's problem [Cat] posed in 1844 is the following: do there exist consecutive positive integers other than 8 and 9 that are both pure powers? This corresponds to the exponential Diophantine equation

$$x^m - y^n = 1.$$

Although this problem was solved completely in the negative by Mihăilescu in 2002 (see Chapter 16), it is still interesting to study the application of Baker's

theory to this problem (even if Baker's theory is not used in Mihăilescu's latest proof, which we give in Chapter 16). In 1976, Tijdeman [Tij] proved that Catalan's problem is a "finite" problem, and we will sketch his proof.

**Theorem 12.9.1 (Tijdeman).** *Let  $x, y, m \geq 2$ , and  $n \geq 2$  be strictly positive integers such that  $x^m - y^n = 1$ . There exists an effectively computable, absolute constant  $C$  such that  $\max\{x, y, m, n\} < C$ .*

*Proof.* By the results of Lebesgue and Ko Chao (Proposition 6.7.12 and Theorem 6.11.8) we may assume that  $m$  and  $n$  are odd. We consider the equation

$$x^m - y^n = \varepsilon ,$$

where  $\varepsilon = \pm 1$  and  $x, y, m, n$  are strictly positive integers with  $n > m > 2$ .

Since

$$(c^n - 1)/(c - 1) = n + (c - 1) \sum_{1 \leq k \leq n-1} \binom{n}{k+1} (c - 1)^{k-1}$$

it follows that  $\gcd((c^n - 1)/(c - 1), c - 1) \mid n$ . We thus have the relations

$$x - \varepsilon = u^n/m^* \quad \text{and} \quad y + \varepsilon = v^m/n^* ,$$

where  $u$  and  $v$  are integers,  $|u|, |v| > 1$ , and where  $m^*$  (respectively  $n^*$ ) is a divisor of  $m$  (respectively of  $n$ ).

It follows from the assumption  $n > m$  that  $x > y$ . Throughout the proof, the constants implied by  $\ll$  and  $\gg$  are absolute.

Consider now the following two linear forms in logarithms:

$$\Lambda_6 := n \log(yu^{-m}) + m \log m^*$$

and

$$\Lambda_7 := mn \log(u/v) - m \log m^* + n \log n^* .$$

Since

$$\left| \frac{y^n}{u^{mn} (m^*)^{-m}} - 1 \right| = \left| \frac{x^m - \varepsilon}{(x - \varepsilon)^m} - 1 \right| \ll \frac{m}{x} ,$$

we note that  $|\Lambda_6| \ll m/x$ , so that using Theorem 12.1.2 and the upper bound  $y \leq 2u^m$ , we obtain

$$\log x \ll m(\log m)(\log n)(\log u) ,$$

hence,

$$n \ll m(\log m)(\log n). \quad (9.1)$$

Furthermore, it follows from

$$\left| \frac{u^{mn}}{(m^*)^m} \frac{(n^*)^n}{v^{mn}} - 1 \right| = \left| \frac{(x - \varepsilon)^m}{(y + \varepsilon)^n} - 1 \right| \ll \frac{n}{y}$$

that  $|\Lambda_7| \ll n/y$ . Thus, we deduce from  $u \leq 3v$  and Theorem 12.1.2 that

$$\log y \ll (\log mn)(\log m)(\log n)(\log v),$$

hence that

$$m \ll (\log m)(\log n)^2. \quad (9.2)$$

Combining equations (9.1) and (9.2) we find that  $n$  is bounded by an absolute constant. It then follows from (9.2) that  $m$  is also bounded by an absolute constant. The fact that  $x$  and  $y$  are also bounded follows from general results on superelliptic Diophantine equations given below in Section 12.11.  $\square$

## 12.10 Thue Equations

### 12.10.1 The Main Theorem

Let  $\mathbb{K}$  be an algebraic number field of degree  $d$ , let  $\alpha_1, \dots, \alpha_n$  be  $n \geq 3$  distinct algebraic integers in  $\mathbb{K}$ , and let  $a$  and  $m$  be nonzero integers. We have the following result (we refer to [Bug-Gyo] for a general totally explicit statement, and to [Bug-Gyo, Sho-Tij] for an extensive list of bibliographic references).

**Theorem 12.10.1.** *With the above notation, the equation*

$$a(x - \alpha_1 y) \cdots (x - \alpha_n y) = m$$

*has only finitely many solutions in integers  $x$  and  $y$ , and all of these can be effectively determined. Moreover, the bound for  $\max\{|x|, |y|\}$  is polynomial in  $|m|$ .*

*Proof.* Let  $x$  and  $y$  be integers satisfying the above equation. For the sake of simplicity assume that  $a = 1$ . Without loss of generality, we may assume that  $x$  and  $y$  are very large, and that  $x/y$  is very close to  $\alpha_1$ . More precisely, setting  $X := \max\{|x|, |y|\}$ , we assume that

$$|x - \alpha_1 y| \ll X^{-n+1} \quad \text{and} \quad |x - \alpha_i y| \gg X, \quad \text{for } i = 2, \dots, n. \quad (10.1)$$

Here and throughout the proof, the constants implied by  $\ll$  and  $\gg$  are effectively computable, and depend only on  $\alpha_1, \dots, \alpha_n$  and  $\mathbb{K}$ . Some of them will be made explicit in the section devoted to algorithmic aspects. Set

$$\beta_i = x - \alpha_i y \quad (1 \leq i \leq 3),$$

and note that

$$(\alpha_1 - \alpha_2)\beta_3 + (\alpha_2 - \alpha_3)\beta_1 + (\alpha_3 - \alpha_1)\beta_2 = 0.$$

Consider now the very small “linear form”

$$\Lambda_8^* := \frac{\alpha_2 - \alpha_3}{\alpha_3 - \alpha_1} \cdot \frac{x - \alpha_1 y}{x - \alpha_2 y} = \frac{x - \alpha_3 y}{x - \alpha_2 y} \cdot \frac{\alpha_2 - \alpha_1}{\alpha_3 - \alpha_1} - 1 .$$

Let  $r$  be the rank of the group of units of  $\mathbb{K} := \mathbb{Q}(\alpha_1)$  and let  $\varepsilon_{1,1}, \dots, \varepsilon_{1,r}$  be a system of fundamental units of  $\mathbb{K}$ . Denote by  $\varepsilon_{2,1}, \dots, \varepsilon_{2,r}$  and  $\varepsilon_{3,1}, \dots, \varepsilon_{3,r}$  the conjugates of  $\varepsilon_{1,1}, \dots, \varepsilon_{1,r}$  in  $\mathbb{Q}(\alpha_2)$  and  $\mathbb{Q}(\alpha_3)$ , respectively, which all belong to the Galois closure  $\mathbb{L}$  of  $\mathbb{K}$ .

There exist an algebraic integer  $\gamma_1$  in  $\mathbb{K}$  of norm at most  $|m|$ , and integers  $b_1, \dots, b_r$  such that  $x - \alpha_1 y = \gamma_1 \varepsilon_{1,1}^{b_1} \cdots \varepsilon_{1,r}^{b_r}$ . We thus have

$$\Lambda_8^* = \left( \frac{\varepsilon_{3,1}}{\varepsilon_{2,1}} \right)^{b_1} \cdots \left( \frac{\varepsilon_{3,r}}{\varepsilon_{2,r}} \right)^{b_r} \frac{\gamma_3(\alpha_2 - \alpha_1)}{\gamma_2(\alpha_3 - \alpha_1)} - 1 ,$$

where  $\gamma_j$  denotes the conjugate of  $\gamma_1$  in  $\mathbb{Q}(\alpha_j)$ , for  $j = 2, 3$ . We then note that

$$B := \max\{|b_1|, \dots, |b_r|\} \ll h(x - \alpha_1 y) \ll \log X . \quad (10.2)$$

Write  $\nu := \gamma_3(\alpha_2 - \alpha_1)/(\gamma_2(\alpha_3 - \alpha_1))$  and note that  $h'(\nu) \ll \log M$ , where  $M := |m|$ . Theorem 12.1.2 gives

$$\log \Lambda_8^* \gg -(\log M)(\log B) ,$$

while our assumptions (10.1) imply that

$$\log \Lambda_8^* \ll -\log X .$$

Combining (10.2) with both estimates, we obtain

$$\log X \ll (\log M)(\log \log X) .$$

We conclude that  $X$  is bounded, but this is not sufficient to obtain a bound that is polynomial in  $M$ .

For establishing the last statement of the theorem it is crucial to apply the improvement quoted just after Theorem 12.1.2, in other words to replace  $B$  by  $B'$ . This gives

$$\log \Lambda_8^* \gg -h'(\nu) \log \left( \frac{B}{h'(\nu)} \right) ,$$

so that we obtain

$$\left( \frac{\log X}{h'(\nu)} \right) \ll \log \left( \frac{\log X}{h'(\nu)} \right) .$$

From this, we deduce that

$$\log X \ll h'(\nu) \ll \log M ,$$

and that there exists a (very small) numerical constant  $\tau$  such that

$$\max\{|x|, |y|\} \leq M^{1/\tau}. \quad (10.3)$$

Thus we have proved not only that  $|x|$  and  $|y|$  are effectively bounded, but also that the bound is polynomial in  $M$ .  $\square$

As a by-product of an explicit version of Theorem 12.10.1, we obtain an effective improvement of Liouville's inequality. Let  $\alpha$  be an algebraic number of degree  $n \geq 3$ . As proved by Liouville, there exists a positive constant  $c_{13}(\alpha)$  such that

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c_{13}(\alpha)}{q^n}, \quad (10.4)$$

for any rational number  $p/q$ . More than one century afterward, Roth established that for any  $\varepsilon > 0$ , there exists a positive constant  $c_{14}(\alpha, \varepsilon)$  such that

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c_{14}(\alpha, \varepsilon)}{q^{2+\varepsilon}},$$

for any rational number  $p/q$ . However, Roth's proof does not yield an explicit value for the constant  $c_{14}(\alpha, \varepsilon)$ . A challenging open problem in Diophantine approximation is to establish an effective version of Roth's theorem. At present, the best that we can do, thanks to (10.3), is to improve only slightly on Liouville's statement, as we now explain.

Let us apply Theorem 12.10.1 with the  $\alpha_i$ 's equal to the complex conjugates of  $\alpha$ , with  $\alpha_1 = \alpha$ . Let  $x$  and  $y$  be integers such that  $x \neq \alpha_i y$  for any  $i = 1, \dots, n$ , and again set  $X := \max\{|x|, |y|\}$ . We then obtain from (10.3) that

$$|x - \alpha y| = \frac{|m|}{|x - \alpha_2 y| \cdots |x - \alpha_n y|} \gg X^{-(n-1)+\tau}.$$

We thus obtain an estimate of the shape (10.4), with the exponent  $n$  replaced by  $n - \tau$ , where  $\tau$  is strictly positive, but very small. Notice that an alternative proof of this result, independent of Baker's theory, has been given by Bombieri [Bom].

### 12.10.2 Algorithmic Aspects

We are in a situation where obtaining an algorithm is possible, since in the linear form  $\log \Lambda_8$ , only the coefficients  $b_i$  are unknown. We need, however, to make explicit the constants in the estimate

$$\log \Lambda_8^* \ll -B, \quad \log \Lambda_8 \ll -B,$$

in order to be able to compute an explicit upper bound for  $B$ , and use the reduction process. This is the purpose of the following lemma. We also derive

an explicit version of (10.1), which may be needed for efficient final enumeration.

In this subsection we restrict to the irreducible case that the left-hand side of the equation is a monic irreducible polynomial in  $\mathbb{Z}[X, Y]$ .

Before stating the lemma, note that the constants derived depend on two choices, the choice of a conjugate  $\alpha_1$  of a root of  $P(X, 1)$  (in practice, a real conjugate; see the end of the lemma) and of an algebraic integer  $\gamma_1 \in \mathbb{Q}(\alpha_1)$  of norm  $m$ . From an algorithmic point of view, we will have to loop over all possible choices.

**Lemma 12.10.2.** *Let  $(x, y)$  be a solution of the Thue equation of Theorem 12.10.1, and assume that  $(x - \alpha_1 y)/\gamma_1$  is a unit of  $\mathbb{Q}(\alpha_1)$  and  $|x - \alpha_1 y| = \min_{1 \leq i \leq n} |x - \alpha_i y|$ . Let  $M = (m_{ij})$  be the inverse of the matrix  $(\log |\varepsilon_{i+1, j}|)_{\substack{1 \leq i \leq r \\ 1 \leq j \leq r}}$ , and define*

$$\lambda_0 = \frac{2^{n-1}|m|}{|g'(\alpha_1)|}, \quad \lambda_1 = 2 \left| \frac{\alpha_2 - \alpha_3}{(\alpha_3 - \alpha_1)(\alpha_1 - \alpha_2)} \right|, \quad \lambda_2 = \min_{1 \leq i \leq r} \frac{n}{\left| \sum_{j=1}^r m_{ij} \right|},$$

$$\lambda_3 = \log \lambda_1 + \lambda_2 \max_{1 \leq i \leq r} \left( \log(3/2) \sum_{j=1}^r |m_{ij}| + \left| \sum_{j=1}^r m_{ij} \log \left| \frac{\alpha_1 - \alpha_{j+1}}{\gamma_{j+1}} \right| \right| \right),$$

$$Y_0 = \max_{1 \leq i \leq n-1} \left( \frac{2\lambda_0}{|\alpha_1 - \alpha_{i+1}|} \right)^{1/n}.$$

Then, if  $|y| > Y_0$ , we have

$$|x - \alpha_1 y| \leq \lambda_0 |y|^{1-n}$$

and

$$\log |\Lambda_8^*| \leq -n \log |y| + \log \lambda_1 \leq -\lambda_2 B + \lambda_3.$$

*Proof.* For  $j \neq 1$  we have

$$2|x - \alpha_j y| \geq |x - \alpha_j y| + |x - \alpha_1 y| \geq |y| |\alpha_1 - \alpha_j|. \quad (10.5)$$

Thus,

$$\begin{aligned} |x - \alpha_1 y| &= \frac{|m|}{\prod_{2 \leq i \leq n} |x - \alpha_i y|} \\ &\leq |m| |y|^{-n+1} \frac{2^{n-1}}{\prod_{2 \leq i \leq n} |\alpha_i - \alpha_1|} = |m| |y|^{-n+1} \frac{2^{n-1}}{|g'(\alpha_1)|}, \end{aligned}$$

which proves the first assertion.

Further, (10.5) shows that for  $|y| \geq Y_0$ ,

$$|\Lambda_8^*| = \left| \frac{\alpha_2 - \alpha_3}{\alpha_3 - \alpha_1} \cdot \frac{x - \alpha_1 y}{x - \alpha_2 y} \right| \leq 2 \left| \frac{\alpha_2 - \alpha_3}{(\alpha_3 - \alpha_1)(\alpha_1 - \alpha_2)} \right| |y|^{-n}.$$

We now need to compare  $B$  and  $\log |y|$ . For this, write

$$\log \left| \frac{x - \alpha_{i+1}y}{\gamma_{i+1}} \right| = \sum_{j=1}^r b_j \log |\varepsilon_{i+1,j}|, \quad 1 \leq i \leq r.$$

From this and the definition of  $M$ , we deduce that

$$b_i = \sum_{j=1}^r m_{ij} \log \left| \frac{x - \alpha_{j+1}y}{\gamma_{j+1}} \right|.$$

Hence,

$$|b_j| \leq \left| \sum_{i=1}^r m_{ij} \log |y| + \sum_{i=1}^r m_{ij} \log \left| \frac{\alpha_1 - \alpha_{i+1}}{\gamma_{i+1}} \right| \right| + \left| \sum_{i=1}^r m_{ij} \log \left| \frac{x - \alpha_{i+1}y}{y(\alpha_1 - \alpha_{i+1})} \right| \right|.$$

Finally, note that

$$\left| \frac{x - \alpha_{i+1}y}{y(\alpha_1 - \alpha_{i+1})} \right| = \left| 1 - \frac{x - \alpha_1 y}{y(\alpha_1 - \alpha_{i+1})} \right| \leq 3/2, \quad \text{for } |y| \geq Y_0,$$

finishing the proof.  $\square$

A comparison with Matveev's bound (note that  $\Lambda_8^*$  is nonzero under our assumptions) then yields an explicit upper bound for  $B$ , which can then be used to initiate the reduction process.

Finally, note the following additional consequences of the lemma:

– For

$$|y| > Y_1 := \max \left( Y_0, \left( \frac{\lambda_0}{\min_{s+1 \leq k \leq s+t} |\operatorname{Im}(\alpha_k)|} \right)^{1/n} \right),$$

we have  $\alpha_1 \in \mathbb{R}$ . This can be used to solve very efficiently totally imaginary Thue equations, and otherwise to restrict the set of roots over which one has to loop. Hence, we can assume that the field  $\mathbb{K}$  has at least one real embedding, so that the only roots of unity are  $\pm 1$ .

– For  $|y| > Y_2 := (2\lambda_0)^{1/(n-2)}$ ,  $x/y$  is a convergent of the continued fraction expansion of  $x/y$ ; this can be used for an efficient final enumeration of “medium-sized” solutions, as in the case of simultaneous Pell equations.

This leads to the following algorithm:

**Algorithm 12.10.3 (Solve a Thue equation)** Let  $P$  be an irreducible monic polynomial with rational integer coefficients, of degree  $d \geq 3$ . This algorithm computes all the solutions of the Thue equation  $Y^n P(X/Y) = m$ .



1. [Algebraic precomputations] Compute the roots  $\alpha_1, \dots, \alpha_n$  of  $P$  (ordered so that the first  $s$  are real, and  $\alpha_{s+t+i} = \overline{\alpha_{s+i}}$ ,  $1 \leq i \leq t$ ), a system of fundamental units  $\varepsilon_{1,1}, \dots, \varepsilon_{1,r}$  of  $\mathbb{Q}(\alpha_1)$ , and a set  $\Gamma$  of nonassociate solutions of the norm equation  $N_{\mathbb{Q}(\alpha_1)/\mathbb{Q}}(u) = m$ . Set  $Y = 0$ .

The following Steps 2–7 should be done for all values of  $(\alpha, \gamma) \in \{\alpha_1, \dots, \alpha_s\} \times \Gamma$ . In the following, the roots should be reordered so that the root under consideration is  $\alpha_1$ .

2. [Computation of various constants] Compute  $\lambda_0, \lambda_1, \lambda_2, \lambda_3, Y_0, Y_1, Y_2, Y = \max(Y, Y_0, Y_1, Y_2)$ .
3. [Computation of Baker's bound] Compute the corresponding Matveev's constant

$$\kappa = 3 \cdot 30^{r+4} (r+1)^{5.5} (n(n-1))^2 h'(\varepsilon_1) \cdots h'(\varepsilon_r) \cdot \log(2\pi) (1 + \log(n(n-1))),$$

and deduce Baker's bound

$$B = \frac{2}{\lambda_2} \left( \kappa \log \frac{\kappa}{\lambda_2} + \lambda_3 + \kappa(1 + \log(r+1)) \right).$$

4. [Reduction, preparation] Let  $\ell = \max(2, r)$ . Compute with sufficient precision approximations of the vector  $(\delta_i)_{1 \leq i \leq \ell}$  defined by

$$\delta_i = \begin{cases} \log |\varepsilon_{2,i}/\varepsilon_{3,i}| & \text{when } r > 1, \\ \text{Arg}(\varepsilon_{2,1}/\varepsilon_{3,1}) & \text{when } r = 1 \text{ and } i = 1, \\ 2\pi & \text{when } r = 1 \text{ and } i = 2, \end{cases}$$

and  $\theta$  defined by

$$\theta = \begin{cases} \log |\gamma_3(\alpha_2 - \alpha_1)/(\gamma_2(\alpha_3 - \alpha_1))| & \text{when } r > 1, \\ \text{Arg}(\gamma_3(\alpha_2 - \alpha_1)/(\gamma_2(\alpha_3 - \alpha_1))) & \text{when } r = 1. \end{cases}$$

5. [Reduction, choice of the parameter] Set  $C$  somewhat larger than  $B^r$ , say  $C = 100B^r$ . If  $\delta_i$  and  $\theta$  are not computed with enough precision to determine exactly  $\lfloor C\delta_i \rfloor$ , increase the working precision and go back to Step 4.
6. [Reduction step] Define

$$G_0 = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ \lfloor C\delta_1 \rfloor & \lfloor C\delta_2 \rfloor & \cdots & \lfloor C\delta_{\ell-1} \rfloor & \lfloor C\delta_\ell \rfloor \end{pmatrix}, \quad v_0 = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \lfloor B\theta \rfloor \end{pmatrix}.$$

Let  $G$  be the matrix of the LLL reduced basis of the lattice generated by the columns of  $G_0$ , and  $v = G^{-1}v_0$ .

Let  $i$  be the smallest index such that  $v_i \notin \mathbb{Z}$ . Set  $d_0 = 2^{(1-r)/2} d(v_i, \mathbb{Z}) \|g_1\|_2$ , where  $g_1$  is the first column vector of  $G$ . If  $d_0 < \sqrt{(r^2/4 + (r-1))B}$ , set  $C \leftarrow 100C$  and go to Step 5. Otherwise, set

$$d'_0 = \sqrt{d_0^2 - (r-1)B - rB/2} \quad \text{and} \quad B' = \frac{1}{\lambda_2} \left( \lambda_3 + \log \frac{2C \log 2}{d'_0} \right).$$

If  $\lfloor B' \rfloor < B$ , let  $B \leftarrow \lfloor B' \rfloor$  and go to Step 5; otherwise, set

$$Y' \leftarrow \max \left( Y', \left( \frac{2\lambda_1 C \log 2}{d'_0} \right)^{1/n} \right).$$

7. [Medium solutions] Compute the convergents  $p/q$  of  $\alpha_1, \dots, \alpha_s$  with  $Y \leq q \leq Y'$ ; if  $F(p, q) = m$ , add  $(p, q)$  to  $\mathcal{S}$ . If  $F(-p, -q) = m$ , add  $(-p, -q)$  to  $\mathcal{S}$ .
8. [Small solutions] For all  $y$  with  $|y| \leq Y$ , find the integer roots of the polynomial  $F(X, y) - m$  (e.g., by computing roots modulo  $p$  and Hensel lifting), and add the corresponding solutions  $(x, y)$  to  $\mathcal{S}$ .
9. [Terminate] Return  $\mathcal{S}$ .

**Remarks.** (1) This algorithm follows rather closely the presentation by Tzanakis and de Weger [Tza-Weg]. It can be optimized in many ways, but should perform quite well as it stands for equations of small degree. A few phenomena can be encountered in small degree (especially in degree 3) that induce a failure of the algorithm: there can be a relation between the  $\delta_i$  (in which case this relation should be taken into account to eliminate one of the variables); or similarly  $\log |\gamma_3(\alpha_2 - \alpha_1)/(\gamma_2(\alpha_3 - \alpha_1))|$  is 0, in which case the homogeneous version of the reduction should be used.

- (2) As in the other algorithms, for the solution to be rigorous, the value of the integers  $\lfloor B\delta_i \rfloor$  should be known exactly. This implies either a careful error analysis in the computations or the use of an interval arithmetic package.
- (3) Various ideas can be used to take into account the fact that the choice of the second and third conjugates to build the linear form is somewhat arbitrary. For instance, Bilu and Hanrot [Bil-Han] have shown how one can use all the conjugates to build a set of  $r-1$  simultaneously small linear forms in  $r$  indeterminates, which, by linear algebra, can be used to build a linear form (the coefficients of which are no longer a priori logarithms of algebraic numbers) in two variables that is very small. The last two parts (reduction, enumeration) become then very similar to the case of simultaneous Pell equations.

## 12.11 Other Classical Diophantine Equations

We begin by stating a special case of Theorem 12.10.1.

**Theorem 12.11.1.** *Let  $F(X, Y)$  be an irreducible, homogeneous, binary integral form of degree  $n \geq 3$ , and let  $b$  be a nonzero integer. The equation*

$$F(x, y) = b, \quad \text{with } x, y \text{ in } \mathbb{Z}, \quad (11.1)$$

*has only finitely many solutions, and all of them can be effectively determined.*

An ineffective version of this result goes back to Thue, and equation (11.1) is commonly called a *Thue equation*. Baker's theory also yields an effective version of a result of Siegel from the 1930s (we refer to [Sho-Tij, Bug] for bibliographical references and for a more general statement; see also Theorem 8.1.2):

**Theorem 12.11.2.** *Let  $f(X)$  be an irreducible polynomial in  $\mathbb{Z}[X]$  of degree  $n \geq 3$  (respectively  $\geq 2$ ), and let  $m \geq 2$  (respectively  $m \geq 3$ ) be an integer. The equation*

$$f(x) = y^m, \quad \text{with } x, y \text{ in } \mathbb{Z},$$

*has only finitely many solutions, and all of them can be effectively determined.*

As already shown by Tijdeman's theorem, the theory of linear forms in logarithms is sufficiently powerful to deal with Diophantine equations in which the exponents are unknown. A general result, due to Schinzel and Tijdeman, is the following.

**Theorem 12.11.3.** *Let  $f(X)$  be an irreducible polynomial in  $\mathbb{Z}[X]$  of degree  $n \geq 3$ . The equation*

$$f(x) = y^z, \quad \text{with } x, y, z \text{ in } \mathbb{Z}, \text{ where } z \geq 2 \text{ and } |y| \geq 2, \quad (11.2)$$

*has only finitely many solutions, and all of them can be effectively determined.*

*Proof.* We prove this theorem, following Brindza, Evertse, and Györy [Bri-Eve-Gyo]. For the sake of simplicity, we will assume that  $f(X)$  is monic. Our goal is to bound  $z$  in terms of  $f(X)$ , and then to conclude by applying Theorem 12.11.2.

Let  $D$  be the discriminant of  $f(X)$ , and  $H$  the maximum of the absolute values of its coefficients. Throughout the proof, the numerical constants implied by  $\ll$  depend only on  $f(X)$ .

Let  $(x, y, z)$  be a solution of (11.2). First note that if  $|x| \leq H + 2$  then we have  $2^z \leq (2H + 3)^n$  since the roots  $\alpha_1, \dots, \alpha_n$  of  $f(X)$  are bounded in absolute value by  $H + 1$ . We can therefore assume that  $|x| > H + 2$ , which implies that  $|x - \alpha_j| \geq 1$  and  $|x - \alpha_j| \leq |y|^z$  for any  $j = 1, \dots, n$ .

In the number field  $\mathbb{K} = \mathbb{Q}(\alpha_1)$  the greatest common divisor of the ideals  $(x - \alpha_1)\mathbb{Z}_{\mathbb{K}}$  and  $(x - \alpha_2) \cdots (x - \alpha_n)\mathbb{Z}_{\mathbb{K}}$  divides the ideal  $f'(\alpha_1)\mathbb{Z}_{\mathbb{K}}$ . Since  $\mathcal{N}_{\mathbb{K}/\mathbb{Q}}(f'(\alpha_1)) = \pm D$  there exist integral ideals  $\mathfrak{a}$ ,  $\mathfrak{b}$ , and  $\mathfrak{c}$  of  $\mathbb{K}$  such that

$$\mathcal{N}_{\mathbb{K}/\mathbb{Q}}(\mathfrak{a}) \leq |D|, \quad \mathcal{N}_{\mathbb{K}/\mathbb{Q}}(\mathfrak{b}) \leq |D|, \quad \text{and} \quad \mathfrak{a}(x - \alpha_1)\mathbb{Z}_{\mathbb{K}} = \mathfrak{b}\mathfrak{c}^m.$$

Let  $h$  be the class number of  $\mathbb{K}$ . The last equation implies that

$$\alpha(x - \alpha_1)^h = \varepsilon \beta \gamma^z, \quad (11.3)$$

where  $\alpha$ ,  $\beta$ , and  $\gamma$  are, respectively, generators of the principal ideals  $\mathfrak{a}^h$ ,  $\mathfrak{b}^h$ ,  $\mathfrak{c}^h$ , and  $\varepsilon$  is a unit of  $\mathbb{K}$ . In particular, we may assume that

$$\max\{h(\alpha), h(\beta)\} \ll 1,$$

where  $h(\xi)$  denotes the absolute logarithmic height of the algebraic number  $\xi$  as given by Definition 12.1.1.

For  $i = 1, \dots, n$ , let  $\phi_i$  be the  $\mathbb{Q}$ -automorphism defined on  $\mathbb{K}$  by  $\phi_i(\alpha_1) = \alpha_i$ , and let  $\{\eta_1, \dots, \eta_r\}$  be a system of fundamental units in  $\mathbb{K}$ . Equation (11.3) gives

$$(x - \alpha_i)^h = (\phi_i(\eta_1))^{k_{i,1}} \cdots (\phi_i(\eta_r))^{k_{i,r}} \phi_i(\beta/\alpha) \phi_i(\gamma)^z, \quad i = 1, \dots, n,$$

where the  $k_{i,j}$  are integers such that  $|k_{i,j}| < z$ . Since  $\max_{1 \leq i \leq n} |x - \alpha_i|^h \leq |y|^{zh}$ , we obtain

$$h(\gamma) \ll \log |y|.$$

If necessary after reordering the roots of  $f(X)$ , we may assume that

$$\min_{1 \leq i \neq j \leq n} \left| \frac{\alpha_i - \alpha_j}{x - \alpha_i} \right| = \left| \frac{\alpha_1 - \alpha_2}{x - \alpha_1} \right|.$$

Recalling that

$$\prod_{1 \leq i \neq j \leq n} \left| \frac{\alpha_i - \alpha_j}{x - \alpha_i} \right| = \frac{|D|}{|y|^{z(n-1)}},$$

we obtain

$$\left| \frac{\alpha_1 - \alpha_2}{x - \alpha_1} \right| \leq \frac{|D|}{|y|^{z/n}}.$$

In addition, we may assume that  $|y|^{z/(2n)} > h|D|$ , since otherwise our theorem is proved. Since  $|x - \alpha_2| \leq |x - \alpha_1|$ , we obtain

$$\Lambda_9^* := \left| \left( \frac{x - \alpha_2}{x - \alpha_1} \right)^h - 1 \right| \leq \left| \frac{x - \alpha_2}{x - \alpha_1} - 1 \right| \cdot h \leq \frac{|D|h}{|y|^{z/n}} < |y|^{-z/(2n)}. \quad (11.4)$$

If  $\Lambda_9^* = 0$ , then  $(\alpha_1 - \alpha_2)/(x - \alpha_1)$  is an algebraic integer and (11.2) gives the upper bound  $z \leq n^2 \log_2 |D|$ .

If  $\Lambda_9^* \neq 0$ , we apply Theorem 12.1.2, and we obtain

$$\log \Lambda_9^* \gg -\log |y| \log z. \quad (11.5)$$

Comparing (11.4) and (11.5), we deduce that

$$z \ll \log z.$$

This proves that  $z$  is bounded. It then remains only to apply Theorem 12.11.2 to conclude the proof of our theorem.  $\square$

## 12.12 A Few Words on the Non-Archimedean Case

Shortly after the publication of the first papers by Baker on the theory of linear forms in logarithms, a few papers on non-Archimedean analogues appeared. The problem becomes the following. Let  $p$  be a given prime number. Let  $\alpha_1, \dots, \alpha_n$  be algebraic numbers whose norms are not divisible by  $p$ , and let  $b_1, \dots, b_n$  be integers. We wish to bound from above the  $p$ -adic valuation of

$$\alpha_1^{b_1} \cdots \alpha_n^{b_n} - 1.$$

It turns out that the presently available estimates are of comparable quality to those provided in the Archimedean case by Theorem 12.1.2, apart from a single point: the dependence on the prime  $p$ . Namely, a supplementary factor  $p^D$  appears, which is essentially due to the fact that the  $p$ -adic exponential function has a bounded disk of convergence (Proposition 4.2.10).

An important application of linear forms in  $p$ -adic logarithms concerns the families of Diophantine equations discussed in the previous section. Instead of looking only for *integer* solutions, we can now deal with rational solutions whose denominators are divisible only by prime numbers from a given finite set.

We end this section with a concrete Diophantine equation solved in [Bug-Mig]. It shows that, sometimes, the use of  $p$ -adic logarithmic forms yields better results than the use of Archimedean ones. Consider the equation

$$\frac{10^n - 1}{10 - 1} = y^q, \quad (12.1)$$

corresponding to the search for perfect powers written in base ten with only the digit 1. Rewriting this equation in the form

$$9y^q 10^{-n} - 1 = -10^{-n},$$

we obtain an upper bound for  $q$  by using estimates for linear forms in three logarithms. However, we can also consider this equation as

$$10^n = 9y^q + 1,$$

and consider the 5-adic valuation  $v_5$  of both sides. On the one hand, it is trivially equal to  $n$ . On the other hand, the 5-adic analogue of Baker's theory allows us to bound  $v_5(9y^q + 1)$  from above: this is not greater than  $c(\log y)(\log q)$ , with a reasonably small constant  $c$ , since it is derived from a linear form in only two logarithms. Observing further that  $n \geq q(\log y)/(\log 10)$ , we immediately obtain that

$$q \log y \leq c(\log 10)(\log y)(\log q),$$

and hence an upper bound for  $q$ . This upper bound is much smaller than the one that can be derived from estimates for three Archimedean logarithms. In

practice, we get  $q \leq 2063$ : since  $q$  may obviously be assumed to be prime, we have replaced our equation (12.1) in three unknowns by about three hundred equations in two unknowns. Various methods are then used to prove that (12.1) has no solution with  $n \geq 2$  and  $q \geq 2$ .

**Remark.** Linear forms in non-Archimedean logarithms can be used to extend some of the algorithms presented above to obtain not only integral solutions, but also  $S$ -integral solutions, i.e., solutions with denominators having prime factors in a prescribed finite set.